

LEGEND

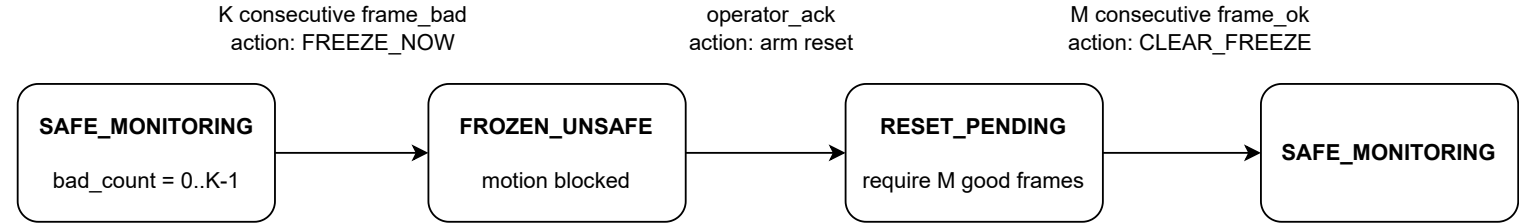
PARAMETERS

K = number of consecutive bad frames required to freeze (e.g., 2)
M = number of consecutive good frames required before re-arm (e.g., 3)

BEHAVIOUR

- **In SAFE:**
 - bad frames increment bad_count
 - if bad_count ≥ K → freeze
- **In FROZEN:**
 - you stay frozen no matter what frames show
 - operator must acknowledge
- **After ack (RESET_PENDING):**
 - require M good frames before allowing SAFE again
 - (prevents immediate unfreeze if marker is still unstable)

GUARDIAN STATE MACHINE.



TRANSITION TABLE				
Current State	Event	Condition	Next State	Action
SAFE_MONITORING	frame_bad	bad_count++; bad_count < K	SAFE_MONITORING	none
SAFE_MONTORING	frame_bad	bad_count reaches K	FROZEN_UNSAFE	FREEZE_NOW
SAFE_MONITORING	frame_ok	reset bad_count	SAFE_MONITORING	none
FROZEN_UNSAFE	operator_ack	-	RESET_PENDING	none
FROZEN_UNSAFE	frame_ok/frame_bad	-	FROZEN_UNSAFE	none
RESET_PENDING	frame_ok	good_count++; good_count < M	RESET_PENDING	none
RESET_PENDING	frame_ok	good_count reaches M	SAFE_MONITORING	CLEAR_FREEZE
RESET_PENDING	frame_bad	reset good_cunt	RESET_PENDING	none